

Ambert Yang Sawaya
Mechanical Engineering Portfolio

TRANSCENDING DISCIPLINES, TRANSFORMING LIVES

Introduction

I am a mechanical engineer with expertise in **robotics, automation, and mechatronics**. My work spans **mechanical design, control systems, and software development**, integrating **Python, MATLAB, CAD, and embedded systems**. Below is a selection of my projects demonstrating my experience in **robotic automation, electrical design, and system modeling**.

- ◆ Key Skills & Expertise:

- Software: MATLAB, Simulink, Python, C++, Arduino, Raspberry Pi
- Hardware: Stepper Motors, Servo Motors, Sensors, PCB Design, Embedded Systems
- CAD & Modeling: SolidWorks, AutoCAD, Creo, Fusion 360
- Control Systems & Automation: GUI Programming, Machine Learning, System Identification

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Mechatronics & Robotics Projects

Multi-Compartmental Lung Tester

Quadrus Medical Inc. & Columbia University (2023 - 2024)

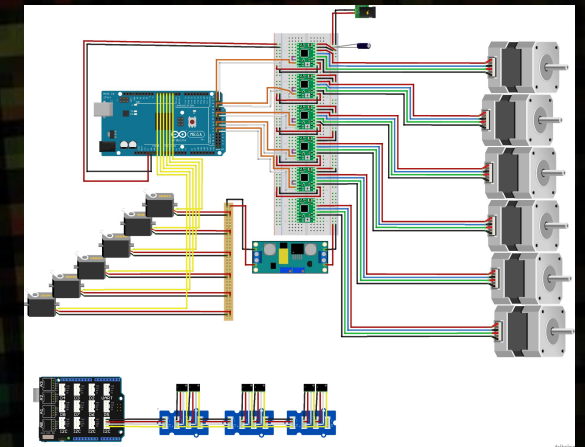
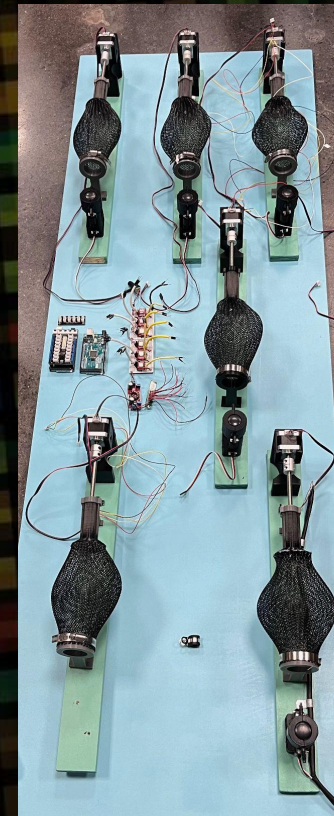
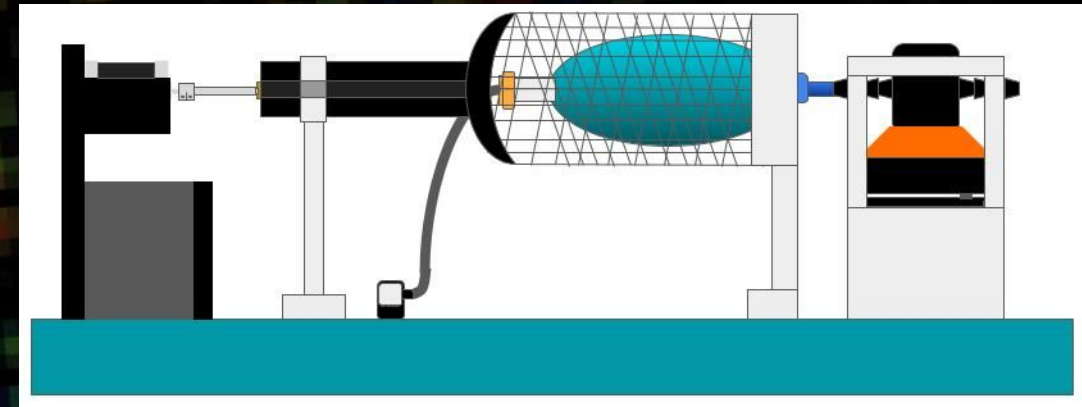
📌 **Goal:** Develop an artificial lung actuated via compliance and resistance, segmented for independent actuation.

📌 Key Contributions:

- Designed the device from **conceptual sketching** to **CAD modeling** and **prototype assembly**.
- Integrated **stepper & servo motors** for actuation and **I2C communication** for real-time sensor feedback.
- Developed a **custom GUI in Python (TKinter)** to control the lung simulation parameters.

📌 **Skills:** Python, Arduino, Mechatronics, 3D Printing, Sensor Integration

📌 **Key Learning:** Advanced **GUI development**, **I2C communication**, and **electromechanical troubleshooting**.



Top: Profile of one compartment.

Bottom: Complete circuit schematic of the project.

CAIR Lab Wireless Haptic Robot

Boston University (Fall 2020 – Spring 2021)

📌 **Goal:** Create wireless haptic modules to assist with haptic sketching for user interaction research.

📌 **Key Contributions:**

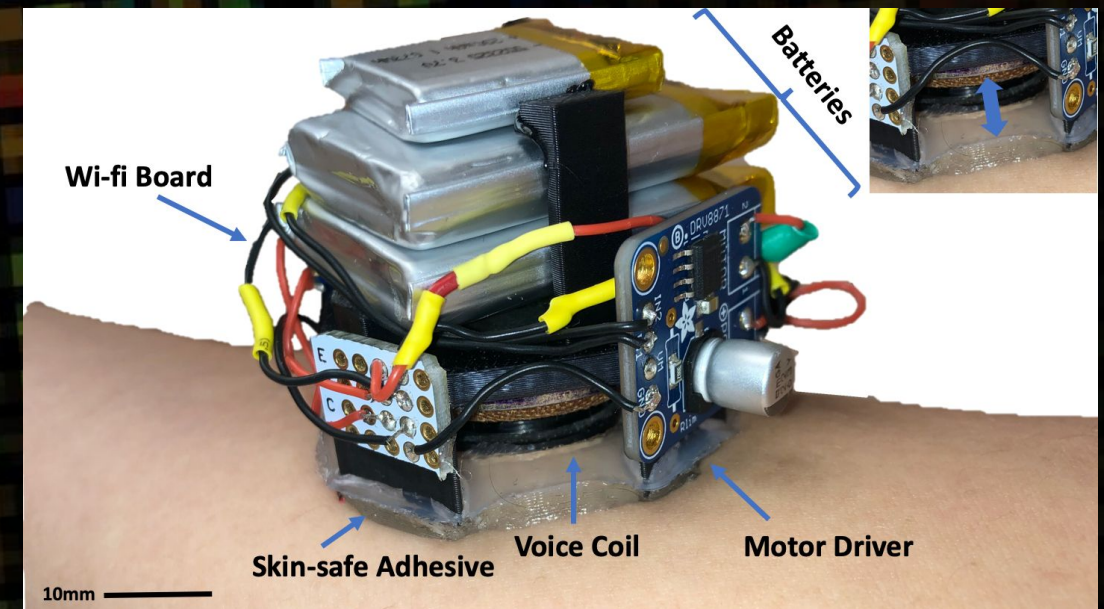
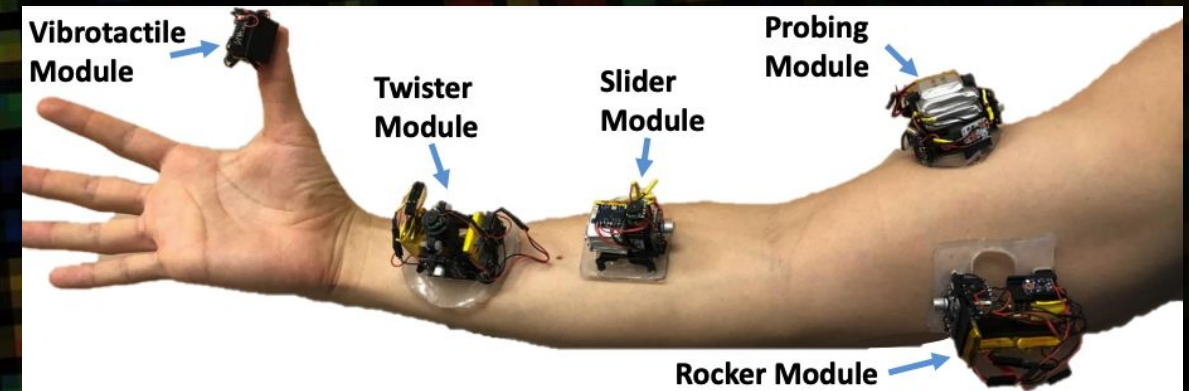
- Designed **CAD models and assembled electronic circuits** for the haptic probing module.
- Assisted in writing a research paper accepted to the **IEEE World Haptics Conference 2021.**

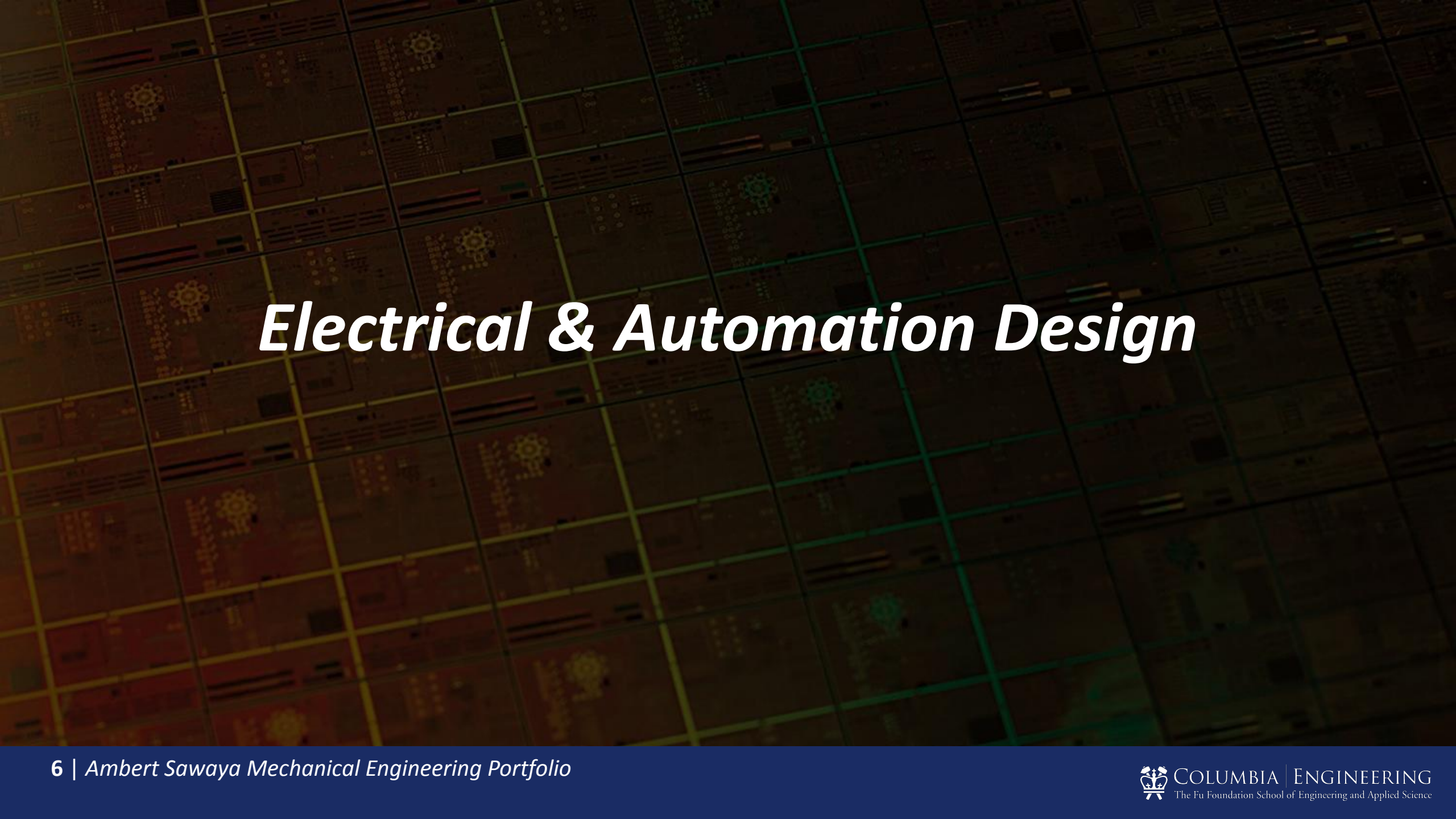
📌 **Skills:** Python, Arduino, Electric Circuits, SolidWorks

📌 **Key Learning:** Soldering, Dremel usage, circuit prototyping, and 3D printing techniques.

Modular Haptic Feedback for Rapid Prototyping of Tactile Displays

Ramón E. Sánchez Cruz^{*1}, Mela C. Coffey^{*1}, Ambert Yang Sawaya¹, and Rebecca P. Khurshid²



A detailed, high-magnification image of a microchip, showing a complex grid of circuitry with various colored lines (yellow, green, black) and small, intricate components. The background is dark, making the circuitry stand out.

Electrical & Automation Design

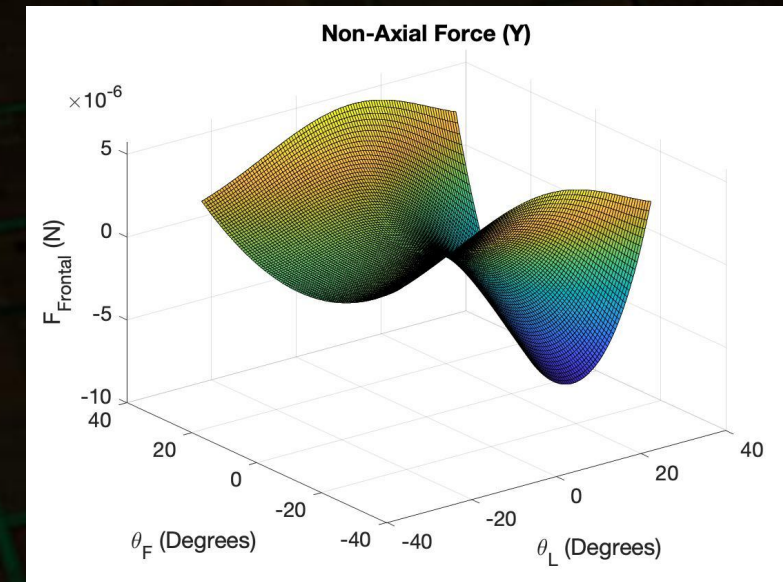
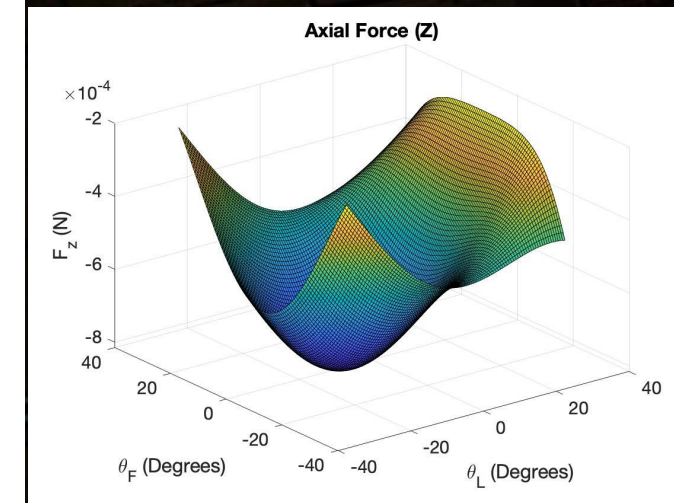
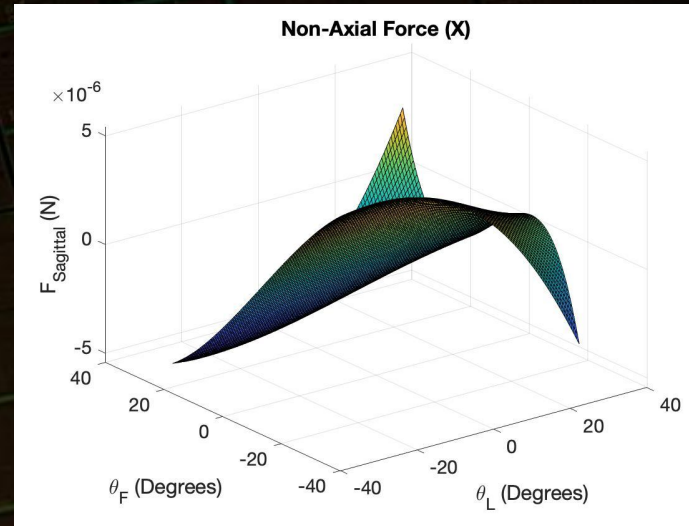
Actuated Gravity Balanced Neck Brace

Columbia University (Spring 2022)

📌 **Goal:** Design a gravity-balanced mechanism for assisted head movement while minimizing neck force.

📌 **Key Contributions:**

- Derived **dynamic equations in MATLAB** and used **Jacobian matrices** to compute required forces.
- Designed a **3D CAD model** with prismatic actuators for movement compensation.
- 📌 **Skills:** Linear Algebra, MATLAB, SolidWorks, Robotic Kinematics
- 📌 **Key Learning:** Advanced kinematics, force minimization strategies, and mechanical system modeling.



Electrical Designer – Soft Gripper

Boston University (Spring 2021)

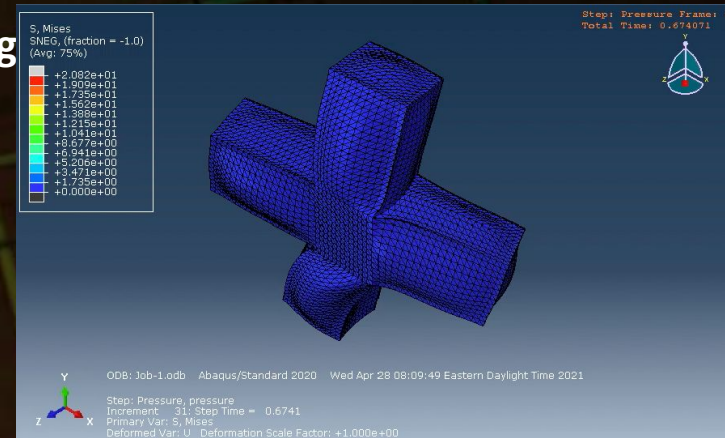
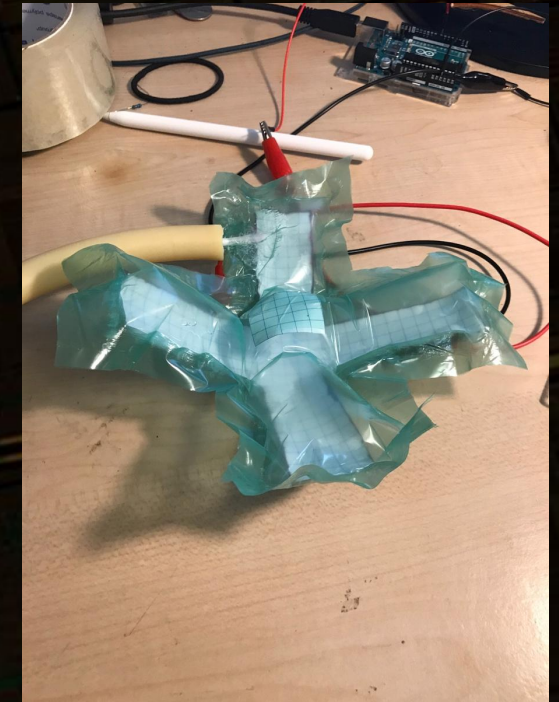
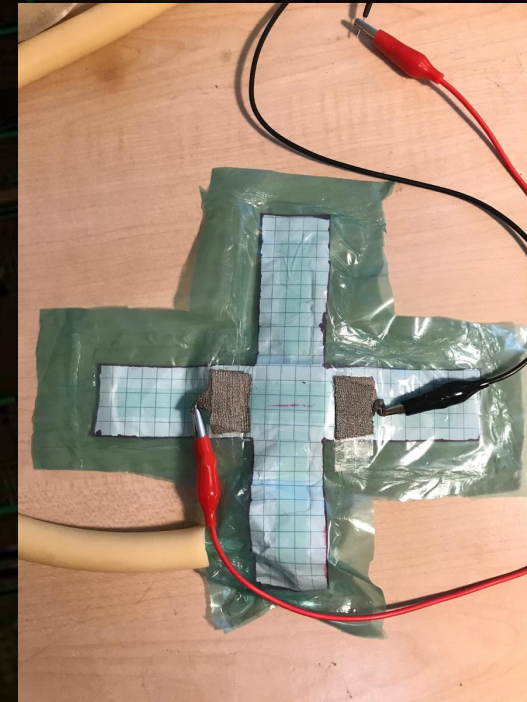
📌 **Goal:** Develop a **lightweight, fluid-actuated robotic gripper** using thermoplastic elastomers (TPE).

📌 **Key Contributions:**

- Designed and fabricated a **4-fingered gripper** with **conductive fabric sensors** for capacitance tracking.
- Modeled inflation behavior using **Finite Element Analysis (FEA) in Abaqus**.

📌 **Skills:** Python, Arduino, Abaqus, SolidWorks, Circuit Design

📌 **Key Learning:** Using **Arduino** for sensor tracking and **Abaqus** for structural simulations.



A detailed, high-magnification image of a microchip, showing a complex grid of circuitry with various colored lines (yellow, green, black) and small components. The text "Other Projects" is centered over this background.

Other Projects

Replication of a Machine Learning Application

Columbia University (Spring 2022)

📌 **Goal:** Reproduce a machine learning model for predicting human migration patterns.

📌 **Key Contributions:**

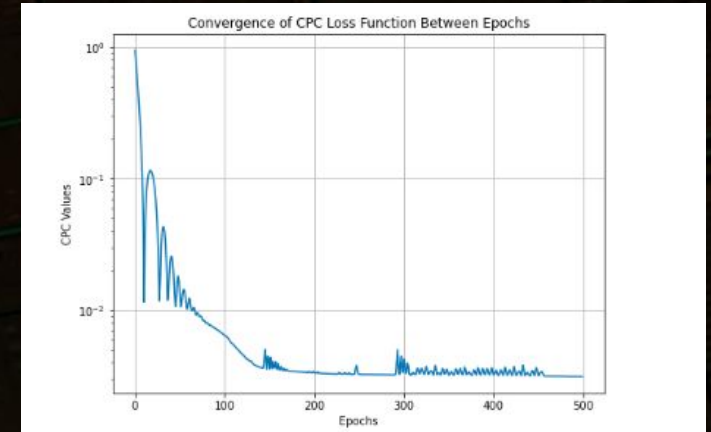
- Processed **IRS Migration Data (2004-2014)** and trained an **Artificial Neural Network (ANN)** using **Python's Keras**.
- Achieved a **1.4% relative error** in predicting migration patterns.

📌 **Skills:** Python, Machine Learning, Data Processing, Neural Networks

📌 **Key Learning:** How to **optimize loss functions** and **fine-tune ANN hyperparameters**.

A Machine Learning Approach to Modeling Human Migration

Caleb Robinson and Bistra Dilkina
Georgia Institute of Technology
School of Computational Science and Engineering
Atlanta, Georgia, 30332
dcrobin@gatech.edu, bdilkina@cc.gatech.edu



```
In [11]: 1 #calculate mean relative error between prediction and validation data
          2
          3 rel_error_array = []
          4 for i in np.arange(len(y_val)):
          5     rel_error = abs(y_val[i]-prediction[i])/y_val[i]
          6     rel_error_array.append(rel_error)
          7
          8 np.mean(rel_error_array, axis=0)

Out[11]: array([0.01457665], dtype=float32)
```

Cooking Robot: ChefServo

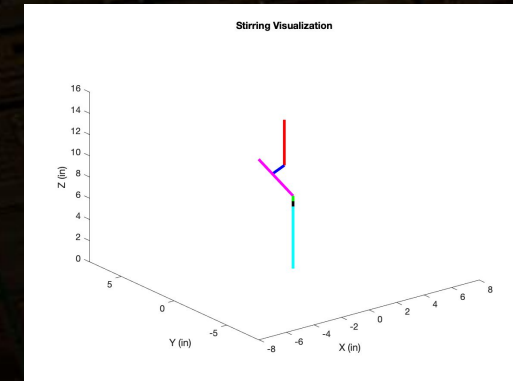
Columbia University (Fall 2021)

📌 **Goal:** Develop a **cooking robot** with automated stirring capabilities.

📌 **Key Contributions:**

- Designed **Denavit-Hartenberg parameters** and implemented **inverse kinematics calculations**.

📌 **Skills:** MATLAB, Robotic Kinematics, Automation



Craigslist Used Car Price Estimator

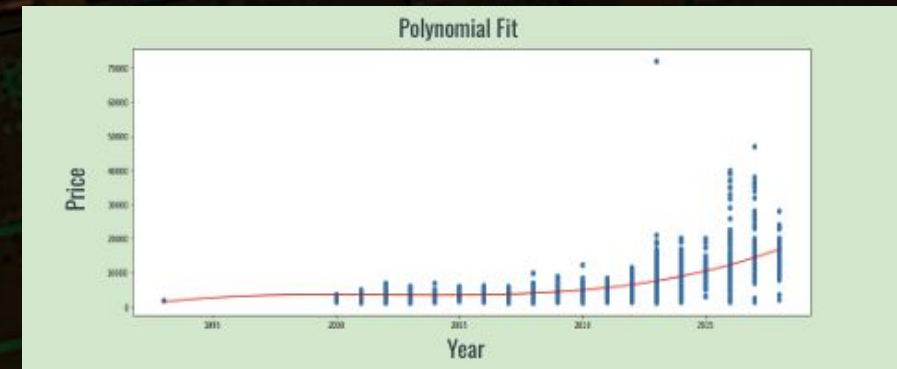
Columbia University (Fall 2021)

📌 **Goal:** Predict **used car prices** based on Craigslist listings using regression models.

📌 **Key Contributions:**

- Assisted in coding **three regression models** to compare data accuracy.

📌 **Skills:** Python, Statistics, Data Science



Enter your car's manufacturer: ford
Enter your car's model: focus
Enter your car's odometer: 10000
Enter your car's year: 2018
Enter your car's condition (New/Excellent/Good/Fair/Salvage): excellent
Enter your city: los angeles
Enter your state (e.g. NY): ca
Enter your zipcode: 90005

Your car will cost approximately \$16030.21